Delta Hedging a Short Straddle and the VaR of the Hedged Book

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1 The Core Idea

A delta-hedged option is not riskless. The hedge removes the first-order exposure to the underlying, and what is left is a bet on volatility: the seller collects implied variance through theta and pays realised variance through gamma. The repository tests this on SPY from 2011 to 2026, then asks a risk question: what is the one-day 99% VaR of a book that has zero delta by construction?

- **Hedging:** a short at-the-money (ATM) straddle is opened every day, hedged daily at the close and held about 30 days.
- **Surface and VaR:** historical option chains are not free, so the daily implied volatility surface is rebuilt from CBOE indices (VIX9D, VIX, VIX3M, SKEW, VVIX). Five VaR methods are then backtested on the hedged book.

2 Why It Matters

- **Losses are convex.** At 20% volatility a 1% daily move earns the short straddle about +0.007 per 100 of spot and a 2% move loses 0.202, thirty times more. Theta pays in small steps and gamma takes it back in large ones.
- **Tails dominate the average.** The mean trade loses 14 bp of strike, but the worst one, opened on 9 March 2020 at 44% implied volatility against 89% realised, lost 10.76%. A tail of that size is invisible in the mean.
- **A VaR blind to gamma and vega says zero.** The delta-normal VaR of a delta-neutral book is zero, while the backtest records a one-day loss of 13.53% (24 March 2020, on a day where the smile fit is degenerate). Any risk system that stops at delta has this hole.
- **The edge depends on the surface.** Selling at the VIX level shows +67 bp per trade, selling at the ATM level shows −14 bp. VIX is a variance swap rate, 3 vol points above the ATM volatility here, so the choice of reference decides whether the trade has an edge.

3 Assumptions

Each line states what the code assumes and which way the result moves if it is wrong.

- **Black-Scholes, constant volatility per trade.** Each straddle is priced and delta-hedged at the ATM implied volatility of the day it is sold, never re-marked. If volatility rises after the trade, the hedge delta is stale and the real hedging error is larger.

- **Daily hedging at the close.** Intraday moves and overnight gaps are not hedged. The gap risk is part of the realised variance, so a real desk hedging more often would lower the error ($1/\sqrt{N}$).
- **SPY as a European index with no dividend.** The code sets $q = 0$ and uses unadjusted closes, so ex-dividend drops count as price moves. The effect is small per day but adds noise to every trade.
- **Rates and calendar.** The rate is the continuously compounded 13-week T-bill from ^IRX, clipped to 0–20%, and time is in calendar days over 365. A weekend is three days of theta.
- **Costs.** 1 bp of notional per stock trade (the opening hedge included) and a half-spread of 0.5 vol point paid once through the straddle's vega. There is no commission, borrow cost or margin. Double the half-spread and the average trade loses 25 bp.
- **Surface shape.** One smile family, $\sigma_{atm}\exp(\beta z + \frac{1}{2}\gamma z^2)$, with β from SKEW and γ from VVIX, the same for all maturities. Real smiles move independently of these indices.
- **Calibration.** The SKEW target is met exactly on 56% of the 3,934 days. On the other 44% the no-arbitrage bound binds and the surface is less skewed than the index says.
- **VaR window.** Historical simulation over 500 days of paired (return, ATM vol change), treated as i.i.d., one-day horizon, a new book each day. After a calm period the window holds no crisis, so the VaR is too low when the crisis starts.
- **Realised P&L.** It is repriced on the reconstructed surface of the next day, the same object the full revaluation methods use. This favours those methods.

4 Method

4.1 P&L of a delta-hedged short option

Each step, a delta-hedged short option earns

$$d\Pi = -\tfrac{1}{2}\,\Gamma S^2\left(\left(\frac{dS}{S}\right)^2 - \sigma^2\,dt\right).$$

The term $\sigma^2 dt$ is the theta income. The squared return is the gamma cost. The trade makes money when realised variance is below implied variance, and the weight ΓS^2 is largest near the strike close to expiry. The code computes this sum along each path (the "forecast") and compares it with the true hedged P&L.

The standard deviation of the hedging error with N rebalances is, to leading order (Derman–Kamal),

$$\text{sd} \approx \sqrt{\frac{\pi}{4}}\,\mathcal{V}\,\frac{\sigma}{\sqrt{N}},$$

where $\mathcal{V}$ is the vega of the option. The error falls with $\sqrt{N}$ and grows with vega, so a long-dated option is harder to hedge in absolute terms. With proportional costs k per side, Leland's adjusted volatility is

$$\sigma_L = \sigma\sqrt{1 + \sqrt{\tfrac{2}{\pi}}\,\frac{2k}{\sigma\sqrt{dt}}},$$

which prices the expected rebalancing cost into the option.

4.2 Volatility surface from CBOE indices

The smile at maturity T is

$$\sigma(K,T) = \sigma_{atm}(T)\,\exp\Big(\beta z + \tfrac{1}{2}\gamma z^2\Big), \qquad z = 3\tanh\Big(\frac{\ln(K/F)}{3\,\sigma_{atm}\sqrt{T}}\Big).$$

- **Level:** $\sigma_{atm}(T)$ interpolates VIX9D, VIX and VIX3M (anchors at 9, 30 and 90 days) linearly in total variance $\sigma^2 T$.
- **Slope:** β is solved so that the risk-neutral skewness of the log return equals $(100 - \text{SKEW})/10$ (median target -3.03, median $\beta = -0.27$). It is capped at the steepest slope that keeps the risk-neutral density non-negative.
- **Curvature:** $\gamma = \text{clip}(0.040 \times \text{VVIX}/0.86,\ 0.01,\ 0.09)$, median 0.043.
- **Rescaling:** VIX is the fair strike of a 30-day variance swap, not the ATM volatility. The whole ATM level is multiplied by a factor λ (median 0.776) so that the smile's own variance swap, $\sqrt{\frac{2e^{rT}}{T}\int \text{OTM}(K)/K^2\,dK}$, equals VIX.

The tanh saturates the moneyness at ± 3 standard deviations. A hard clip would give a negative density at its kink.

4.3 VaR of the hedged book

The book is a short 30-day ATM straddle, hedged to zero delta and rebuilt every day. All methods use the same 500-day window of pairs (log return x, change in ATM vol $\Delta\sigma$), at $\alpha = 99\%$. For each historical pair, with $\Delta S = S_0(e^x - 1)$:

Method	Loss scenario
Delta-normal	$z\,\lvert\Delta_{book}\rvert\,\text{sd}(x)$, which is zero
Delta-gamma-theta	$\frac{1}{2}\Gamma\,\Delta S^2 + \Theta\,\Delta t$
Delta-gamma-theta-vega	same, plus $\mathcal{V}\,\Delta\sigma$
Full revaluation, parallel shift	reprice the straddle at $(S_0 + \Delta S,\ \sigma_{atm} + \Delta\sigma)$ at the old strike
Full revaluation, smile	reprice at the old strike on the smile at the new spot, ATM level shifted by $\Delta\sigma$

Both full revaluation methods also move the maturity from 30 days to 30 days minus the calendar gap to the next date (one day on a normal weekday), and add the P&L of the delta-hedge shares, $\Delta_{straddle}\,\Delta S$. The VaR is the empirical 1% quantile of the loss scenarios, in % of spot. Θ is negative for the long straddle, so the second term is theta income for the short seller and the loss is the opposite sign. The realised P&L of the next day is the book repriced on the next day's smile. Two tests are used: Kupiec (exception count against 1%) and Christoffersen (independence of exceptions). Both are compared with a χ^2_1.

5 Method Comparison

Method	Strength	Weakness
Delta-normal	Cheap, standard for linear books	Zero when net delta is zero
Delta-gamma-theta	Captures the convexity	Ignores vol moves: 107 exceptions
Delta-gamma-theta-vega	Adds the first-order vol term	Linear in $\Delta\sigma$; conservative
Full revaluation, parallel	Exact for the pair, no Greeks	Needs a vol-shift rule; no smile
Full revaluation, smile	Includes skew roll-down	Needs a surface; uses the model it tests

A bank with real quotes would use full revaluation on its own surface and keep the Greek-based methods as a cross-check. The gamma-vega approximation passes without a surface, but its mean VaR (194 bp) is the highest of the three that pass, so the approximation costs capital for the same coverage.

6 Results

Simulation checks. Hedging error with $\sigma = 20\%$, a 30-day ATM call and 20,000 paths:

N	Measured sd	Formula	Gap
5	0.8363	0.9062	-7.7%
21	0.4252	0.4422	-3.9%
250	0.1266	0.1282	-1.2%

The formula is a leading-order result, so the measured error sits below it and the gap shrinks as N grows. A second test sells at 20% while the path has 25% volatility. Hedging at the realised volatility gives the same mean loss as hedging at the implied one (-0.5719 against -0.5681, and the exact price gap is -0.5716) with a much smaller standard deviation (0.1257 against 0.2709). The mean is set by the volatility sold, the spread of outcomes by the volatility hedged. A third test with 10 bp per trade gives a mean P&L of -0.0485 ($N = 21$) and -0.0513 ($N = 100$) at Leland's volatility, close to the cost of the opening hedge (-0.0512 and -0.0513), while selling at 20% would lose 0.1936 and 0.3695. The simulations do not show that real markets behave like geometric Brownian motion. They show that the code hedges correctly when they do.

SPY, 2011 to 2026. One ATM straddle is sold every day and delta hedged at the close, with a half-spread of 0.5 vol point and 1 bp per stock trade:

Metric	Value
Trades (overlapping / non-overlapping)	3,911 / 181
Mean P&L, in bp of strike	-14
Win rate	52.5%
Same trades sold at the VIX level	+67 bp, win rate 80.8%
Mean implied vs realised volatility	14.7% vs 14.7%
R^2 of P&L on the gamma-theta sum	0.934
R^2 of P&L on implied minus realised vol	0.597
Worst trade	opened 9 Mar 2020, -10.76%

The average implied and realised volatilities are equal, so the mean volatility premium on the rescaled ATM level is zero. The median gap is +1.2 vol points in the seller's favour, but the rare spikes take it back. The −14 bp is then costs: with no costs the mean is +0.1 bp (win rate 59.6%), 1 bp per stock trade takes 2.8 bp and the half-spread takes 11.3 bp. The +67 bp comes from pricing the same trades at the VIX level, which sits above the ATM volatility when skew is negative, so it depends on the rescaling and is closer to the variance risk premium of the index than to a tradable edge. With 181 independent trades the mean of −15 bp has a t-statistic of −2.0, so costs are significant, not skill. Results move with the regime: the yearly mean ranges from +32 bp (2012) to −71 bp (2022). Win rate is above 50% but the median trade earns a little and the 1% worst trades lose 3.9% of strike or more, which is the skew of −2.5 in the P&L. The R^2 of 0.93 on the gamma-theta sum says the replication works; the lower 0.60 on implied minus realised volatility says gamma-weighted variance matters more than the average volatility gap.

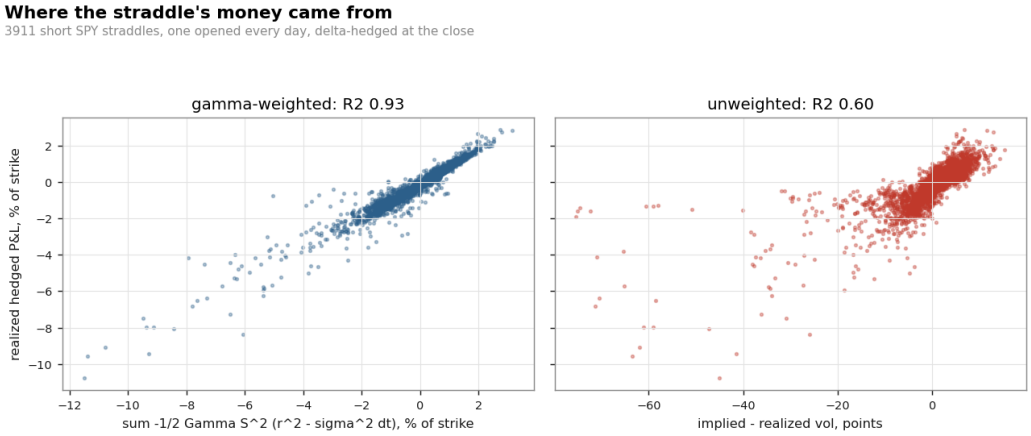


Figure 1: Hedged P&L of the 3,911 short straddles against the gamma-theta sum (left) and against implied minus realised volatility (right).

VaR backtest. 3,433 days from 2 January 2013 to 26 August 2026, 500-day window, 34.3 exceptions expected:

Method	Mean VaR (bp)	Exceptions	Rate (%)	Kupiec p
Delta-normal	0	1,495	43.55	0.000
Delta-gamma-theta	124	107	3.12	0.000
Delta-gamma-theta-vega	194	26	0.76	0.135
Full revaluation, parallel	178	29	0.84	0.348
Full revaluation, smile	158	39	1.14	0.433

- **Delta-normal:** the VaR is zero, so every losing day is an exception. It is a baseline only.
- **Clustering:** the Christoffersen p-value is below 0.001 for all five methods.
- **Ranking:** the order among the last three is not stable. Without the 17 days that touch a degenerate smile, only full revaluation on the smile passes Kupiec ($p = 0.978$; the other two have $p = 0.004$ and 0.025). If the realised P&L is repriced with a parallel vol shift, full revaluation on the smile fails (2.18% exceptions).
- **Worst days:** the largest realised losses (13.53% on 24 March 2020) fall on days where the smile fit is degenerate, so they are model errors as much as market moves.

Dropping the vega term is the largest error: delta-gamma-theta has 107 exceptions against 26 for the same method with vega. The reason is in the data: daily changes in ATM volatility have a standard deviation of 1.6 vol points against 1.08% for the index return, and the correlation between the two is -0.79, so a fall in spot comes with a rise in volatility and a short straddle loses on both. The smile method has a lower mean VaR (158 bp) than the parallel one (178 bp) because, after a fall in spot, the old strike is out of the money and the smile lowers its volatility. Exceptions cluster in time: for the smile method, 9 fall in 2018, 10 in 2020 (all between February and April), 7 in 2015 and 5 in 2025. The VaR is slow: on 24 March 2020 it stood at 3.90% while the loss was 13.53%. The passing of Kupiec does not mean the VaR is adequate in a crisis; it means the average number of breaches is right.

Surface against two real SPX chains (CBOE delayed quotes, 30-day ATM volatility):

Date	Chain ATM	VIX	Rescaled surface
22 Sep 2026	11.3%	14.2%	10.7%
28 Sep 2026	12.7%	16.1%	12.0%

The level is within 0.7 vol point after rescaling, while the raw VIX is 2.9 to 3.4 points too high. The shape is the weak point. On 28 September the rescaled smile is 4.1 vol points above the chain at a strike 10% below the forward (puts too expensive), and 2.5 points below at a strike 5% above it (calls too cheap). The surface is too steep.

7 Worked Example

One day of a hedged short straddle (illustrative): $S = K = 100$, $\sigma = 20\%$, 30 days, $r = 0$, one day $dt = 1/365$. For the call, $\Gamma = 0.06955$ and the straddle has $\Gamma = 2 \times 0.06955 = 0.13910$.

$$
\begin{aligned}
\tfrac{1}{2}\Gamma S^2 &= 0.5 \times 0.13910 \times 10{,}000 = 695.5,\\
\text{theta income} &= 695.5 \times 0.20^2/365 = 0.0762,\\
\text{move } +1\%: &\quad -695.5 \times 0.01^2 + 0.0762 = -0.0696 + 0.0762 = +0.0067,\\
\text{move } +2\%: &\quad -695.5 \times 0.02^2 + 0.0762 = -0.2782 + 0.0762 = -0.2020.
\end{aligned}
$$

The break-even daily move is $\sigma\sqrt{dt} = 20\%/\sqrt{365} = 1.05\%$.

Derman–Kamal for the same option with $N = 21$: the call vega is $\mathcal{V} = 11.43$, so sd $= 0.8862 \times 11.43 \times 0.20/\sqrt{21} = 0.442$, the formula column above.

One VaR scenario (illustrative): the same straddle, a historical pair $x = -2\%$ and $\Delta\sigma = +2$ vol points, so $\Delta S = -1.98$. The straddle vega is $\mathcal{V} = 22.87$ per 1.00 of volatility, theta is -0.0762 a day, the straddle is worth 4.574.

$$
\begin{aligned}
\text{delta-gamma-theta:} &\quad -(0.5 \times 0.1391 \times 1.98^2 - 0.0762) = -(0.2727 - 0.0762) = -0.196,\\
\text{plus vega:} &\quad -0.196 - 22.87 \times 0.02 = -0.654,\\
\text{full revaluation, parallel:} &\quad -0.622.
\end{aligned}
$$

The parallel revaluation reprices the straddle at 29 days, which gives -0.5762, and adds the hedge shares: the straddle delta is 0.02287, so they contribute $0.02287 \times (-1.98) = -0.0453$. The sum is -0.6215, shown as -0.622. As a share of spot the loss is 0.20%, 0.65% and 0.62%. The vega term (0.457) is larger than the gamma loss net of theta (0.196). With the smile (illustrative $\beta = -0.27$, $\gamma = 0.043$, flat 20% term structure), the old strike has a volatility of 20.2% at the new spot instead of 22%, and the loss is only 0.246, or 0.25% of spot.

Kupiec test for full revaluation on the smile: $n = 3433$, $x = 39$, $p = 0.01$, so 34.33 expected and $\hat{p} = 1.136\%$.

$$\begin{aligned}\ln L_0 &= 3394 \ln 0.99 + 39 \ln 0.01 = -213.7125,\\ \ln L_1 &= 3394 \ln(1-\hat{p}) + 39 \ln \hat{p} = -213.4051,\\ LR &= -2\,(-213.7125 + 213.4051) = 0.6147 \;\Rightarrow\; p\text{-value} = 0.433.\end{aligned}$$

Sensitivity.

- *Volatility doubles* (illustrative, $\sigma = 40\%$): the half-gamma term falls from 695.5 to 347.3 and theta income rises to 0.152 a day. The break-even move doubles to 2.09%, a 2% move now earns +0.013 and Derman–Kamal at $N = 21$ is 0.883. The higher implied volatility pays for more movement and also makes the hedge error twice as large.
- *Costs.* The mean trade in bp for a half-spread of 0, 0.25, 0.5 and 1 vol point is −2.6, −8.3, −13.9 and −25.2 (stock cost kept at 1 bp). The win rate falls from 58.1% to 45.8%.
- *VaR window.* With 250 days, 3,683 days are tested and the exceptions are 126, 29, 32 and 49 for delta-gamma-theta, gamma-vega, parallel and smile (Kupiec $p = 0.000$, 0.178, 0.413, 0.055). With 1,000 days, 2,933 days and 64, 24, 28 and 39 (0.000, 0.307, 0.804, 0.088). The mean VaR of the smile method is 137 bp, 158 bp and 153 bp for the 250, 500 and 1,000 day windows. All three windows pass at 5% for the vega and full revaluation methods, with the smile method closest to the line at 250 and 1,000 days.

8 Limits and Extensions

- **Reconstructed smile:** only the ATM level and the index-implied smile are real data. On the checked chains, puts struck below the spot come out up to 4 vol points too expensive and calls struck above it up to 3 points too cheap.
- **Circular backtest:** the realised P&L is repriced on the same reconstructed surface that the full revaluation methods use, which favours them.
- **One book, one product:** the results are for a 30-day ATM straddle on one index. They say little about other maturities or a book with many strikes.

Three extensions would address these limits. Replacing the realised P&L by the P&L of a real straddle, from daily chains, would break the circularity. A filtered historical simulation (rescale the window by current volatility) would make the VaR react faster after calm periods. An ES backtest on the same book would measure how large the exceptions are, which a count of exceptions cannot show.

9 Conclusion

A delta-hedged short option sold at fair implied volatility earns about nothing before costs on SPY, and what is left is exposure to rare volatility spikes that a Greek-based VaR can miss unless it has a vega term. A risk manager should therefore size such a book on a stress scenario with a spot and volatility shock together, and use the VaR only as a daily control.